

TAHA MERGHANI

taha.y.merghani@gmail.com | linkedin.com/in/tahamerghani | github.com/taha-y-merghani | taha-y-merghani.github.io

FORWARD DEPLOYED AI ENGINEER

SUMMARY

Applied AI/NLP engineer with experience building and deploying machine learning systems across Siri tooling, speech recognition, computer vision inspection, customer analytics, and local LLM interfaces. Strong fit for customer-facing AI deployment work: translating ambiguous workflows into working prototypes, integrating messy data, diagnosing model failure modes, and communicating technical tradeoffs clearly. M.S. in Computer Science from Georgia Tech, B.S. in Computer Engineering with 4.0 GPA, and research presented at NeurIPS Black in AI and NAACL workshops.

EXPERIENCE

Independent AI Engineer | Apr 2023 - Present

- Built local voice-interface prototypes using Whisper, Ollama, and Mistral 7B, optimizing for privacy, latency, and execution on constrained consumer hardware.
- Authored technical writing connecting human skill acquisition, AI research, and real-world decision-making, reaching 200k+ readers on r/ArtificialIntelligence.
- Rebuilt public technical portfolio around applied AI systems, NLP research, speech interfaces, and field notes on operational deployment.

Solutions Architect, Mesa Associates | Nov 2022 - Apr 2023

- Deployed computer vision workflows on Oracle Cloud Infrastructure for automated electrical equipment inspection.
- Labeled, trained, and tested models for utility inspection use cases involving substation and field imagery.
- Wrote Python notebooks and scripts to process drone imagery and metadata for downstream inspection workflows.
- Worked at the boundary between AI tooling, operational constraints, and customer-facing engineering needs.

AI Software Engineer, Decooda | Jul 2019 - Aug 2020

- Built ETL pipelines with Matillion and Snowflake to support customer analytics and machine learning workflows.
- Implemented and deployed machine learning solutions using PyTorch and Docker.
- Worked across data engineering, model experimentation, and production-oriented customer analytics systems.

Graduate Research Assistant, Georgia Institute of Technology | Aug 2017 - May 2019

- Researched NLP systems robust to language variation in social media, including stylistic variation in part-of-speech tagging.
- Published work at the NAACL Workshop on Stylistic Variation and the NeurIPS Black in AI Workshop.
- Built models and experiments connecting linguistic variation, social networks, and geospatial language patterns; conducted conversational AI research under Dr. Mark Riedl on collaborative story-generation agents.

Software Engineering Intern, Apple Siri | Summer 2016

- Built Python tooling for debugging and visualizing Siri NLU issues across noisy bilingual bug reports and 40+ languages.
- Integrated custom regex parsing, internal error-management data, and Siri REST APIs into a tool used by 120+ engineers.
- Saved an estimated \$200k+ in engineering time, exceeded project milestones by more than 20%, and was selected as a Distinguished Intern presenter, one of 4 selected from a cohort of 33.

Research Intern, MIT CSAIL | Summer 2015

- Developed lexical and pronunciation-dictionary experiments for Egyptian Arabic automatic speech recognition under Prof. James Glass.
- Built scripts in C++, Bash, and Kaldi to train and compare GMM-HMM and DNN acoustic models on a 10-hour Egyptian Arabic corpus.
- Found that grapheme-based lexicons outperformed diacritized approaches in the low-resource setting, showing how additional linguistic signal can become noise.

EDUCATION

M.S. Computer Science, Georgia Institute of Technology | GPA: 3.87/4.0 | Machine Learning, NLP, Computational Social Science

B.S. Computer Engineering, Jackson State University | GPA: 4.0/4.0 | Jacksonian Award of Distinction, Highest GPA

SELECTED PUBLICATIONS & WRITING

- Balusu, Merghani, Eisenstein. "Stylistic Variation in Social Media Part-of-Speech Tagging." NAACL Workshop, 2018.
- Merghani. "Application of the Hilbert-Schmidt Independence Criterion to Lexical Geographical Variation in Lyon, France." NeurIPS Black in AI Workshop, 2018.
- "From Sudan to Silicon Valley: Beyond the Resume." Shared by Jeff Dean.
- "What 5,000 Hours of Mastering Tekken Taught Me About AI Research." #1 on r/ArtificialIntelligence.

SKILLS

AI/ML: LLMs, NLP, speech recognition, computer vision, PyTorch, Hugging Face, Kaldi, Whisper, Ollama | **Engineering:** Python, SQL, Docker, FastAPI, Jupyter, Bash, C/C++ | **Data & Cloud:** Snowflake, Matillion ETL, Oracle Cloud Infrastructure, Pandas | **Languages:** Arabic native, English near-native